

# Isha Chaudhary

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## Education

**University of Illinois Urbana-Champaign** 2022 - 2026

*Ph.D. in Computer Science; Guide: Prof. Gagandeep Singh*

**University of Illinois Urbana-Champaign** 2022 - 2025

*M.S. in Computer Science with Thesis; Guide: Prof. Gagandeep Singh* GPA: 3.97/4.00

**Indian Institute of Technology Delhi, India** 2018 - 2022

*B.Tech in Electrical Engineering (minor in CS); GPA: 9.3/10.0* Class Rank: 1/51

## Research Interests

**Safety and trustworthiness of frontier AI systems:** certifying catastrophic risks and adversarial robustness with formal guarantees for LLM and agentic system behaviors. <https://certifyllm.com/>

## Selected Publications

- **Isha Chaudhary\***, Vedaant Jain\*, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Revealing Interpretable Failure Modes of VLMs*. Fagen workshop @ ICML 2026. (\* indicates equal contribution)
- **Isha Chaudhary**, Vedaant Jain, Gagandeep Singh. *Certifying Reading Comprehension in Large Language Models*. **AISTATS 2026 Spotlight**, also in SeT LLM Workshop @ ICLR 2024.
- Chengxiao Wang, **Isha Chaudhary**, Qian Hu, Weitong Ruan, Rahul Gupta, Gagandeep Singh. *How Catastrophic is Your LLM? Certifying Risks in Conversation*. **ICLR 2026**.  
**Impact:** Deployed in Amazon's Nova evaluation suite
- **Isha Chaudhary**, Qian Hu, Manoj Kumar, Morteza Ziyadi, Rahul Gupta, Gagandeep Singh. *Quantitative Certification of Bias in Large Language Models*. **ICLR 2025**.
- **Isha Chaudhary**, Vedaant Jain, Avaljot Singh, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. *Lumos: Let there be Language Model System Certification*. PAGe workshop @ PLDI 2026.  
**Impact:** In deployment within National Research Platform's model cards
- Jehyeok Yeon, **Isha Chaudhary**, Gagandeep Singh. *Quantifying Distributional Robustness of Agentic Tool-Selection*. AIWILD @ ICLR 2026.
- Jason Vega\*, **Isha Chaudhary\***, Changming Xu\*, Gagandeep Singh. *Bypassing the Safety Training of Open-Source LLMs with Priming Attacks*. **ICLR 2024** (Tiny Papers track). (\* indicates equal contribution)  
**Impact:** Attack Success Rate  $\uparrow 3.3\times$  over baselines on harmful behavior benchmarks
- **Isha Chaudhary**, Shuyi Lin, Cheng Tan, Gagandeep Singh. *Specification Generation for Neural Networks in Systems*. **ICSE 2026** (Poster), also in ML4Wireless Workshop @ ICML 2025.
- **Isha Chaudhary**, Alex Renda, Charith Mendis, Gagandeep Singh. *COMET: X86 Cost Model Explanation Framework*. **MLSys 2024**, also in XAI-in-Action Workshop @ NeurIPS 2023.

## Experience

**Anthropic** USA

*AI Safety Fellow* Aug 2026 - present

- Stress-testing alignment evaluation of frontier model systems

**Amazon AGI** Boston, USA

*Applied Scientist Intern with Dr. Rahul Gupta and Dr. Weitong Ruan* May 2026 - Aug 2026

- Constitutional AI alignment by adversarial post-training with Petri over policies for frontier agentic system safety

**Amazon Web Services** Seattle, USA

*Applied Scientist Intern with Dr. Ganyu Teng and Dr. Shawn Zhang* Sep 2025 - Dec 2025

- Developed a principled framework to stress-test scalable-oversight-enabled agentic systems via adversarial prompt injections, using CVaR to expose worst-case alignment failures hidden by average-case metrics
- Finetuned Qwen-1.5B as a transferable adversary with GRPO to generate effective prompt-injection attacks against scalable-oversight judge models.

**Microsoft Research** Redmond, USA

*Research Intern with Dr. Ryan Beckett, Dr. Siva Kakarla, Prof. Francis Yan, Dr. Behnaz Arzani* May 2025 - Aug 2025

Developed LLM-powered semantic analysis of system logs for anomaly detection, leveraging frontier model capabilities for safety-critical infrastructure monitoring.

**Microsoft AI (Bing)** Redmond, USA

Developed machine learning models for revenue optimization in Bing search.

### Adobe Research (Cloud Tech group)

Bengaluru, India

Research Intern / Guide: Dr. Gaurav Sinha / **US patent no. US 2023/0274310 A1**

May 2021 - Aug 2021

Multi-task Learning with Label Proportions: Jointly disaggregated multiple aggregate labels with individual features

### Blockchain IoT Security Project: DST-JST Grant on IoT Security

IIT Delhi, India

Research Assistant / Guide: Prof. Subodh Sharma, IIT Delhi / Denso Intl. India

Sept 2020 – May 2021

Designed & developed Hyperledger Fabric-based supply chain system for real-time order traceability & security

## Awards

- Selected for **Y Combinator's** AI Startup School (2026)
- Awarded **Best Presentation Award** (Security and Privacy track) at CSL student conference 2025 (2025)
- Awarded **Institute Silver Medal & Prof. C.S. Jha Memorial Excellence Award** at IIT Delhi (2022)
- Top 7% merit award:** In top 7% high GPA students at IIT Delhi for 5 semesters: semesters 3, 4, 6, 7, 8 (2019-22)
- Design & Innovation Summer Award (MHRD Grant, Govt. of India)** (2019)
- KVPY:** Qualified for the Kishore Vaigyanik Protsahan Yojana Fellowship (IISc); All India Rank 232 (2016)
- NTSE:** Received National Talent Search Scholarship, given to top 1000 students nationwide by NCERT (2015)

## Talks

### Certifying LLMs

- NSF's National AI Research Resource (NAIRR) Pilot 2026
- Microsoft Research Redmond and India, CSL student conference, IIT Delhi 2025
- Amazon-Illinois Center on AI for Interactive Conversational Experiences 2024

### Teaching Smarter with AI: How to Use AI in Classrooms Without Losing Control

- Graduate TA training academy organized by CITL, UIUC 2025

### Exploring COMET: Neural Cost Model Explanation Framework

- Google Compiler Research reading group, VMWare Research 2024

## Teaching

### CS 477: Formal Software Development Methods

UIUC, USA

Guest Lectures on formally certifying frontier models

Apr 2026

### Grad Academy, CITL

UIUC, USA

Instructor: Isha Chaudhary, UIUC

Aug 19, 2025

### Advanced Topics in Programming Systems - Trustworthy AI Systems (CS521)

UIUC, USA

Instructor: Prof. Gagandeep Singh, UIUC

Aug 2024 – Dec 2024

### Calculus (MTL100)

IIT Delhi, India

Instructor: Prof. Sivananthan Sampath, IIT Delhi

Aug 2021 – Dec 2021

## Grants

- Awarded **ICML 2023 Travel Grant** (2023)
- Awarded scholarship for **Programming Languages Mentoring Workshop @ PLDI 2023** (2023)
- NSF Travel Grant for IEEE SaTML 2023** (2022)

## Service

- Area Chair for MLSys 2025.
- Reviewer for NeurIPS 2026; Computing Surveys 2026; ICML 2026; KDD 2026; COLM 2026; ICLR 2025, 2026; FAccT 2025, 2026; MLSys 2025; AIES 2024, 2025; Tiny Papers @ ICLR 2024.
- Artifact Evaluator for VMCAI 2024.
- Main organizer for Formal Methods and Software Engineering seminar, Spring 2024 & Fall 2024, at UIUC.
- Mentor for the Undergraduate Research Apprenticeship Program (URAP) at UIUC, 2024-25
- Mentor in the Undergraduate Research in Scientific Advancement (URSA) program at UIUC, 2024-25
- Mentor in SIGPLAN-M long-term mentorship programme
- Mentor in K-12 Broadening Participation in Computing program at UIUC, Summer 2023.
- (Guided a team of high-school students in successfully learning & conducting research on LLMs.)

## Undergraduate Thesis Project

### Security & Privacy of SCADA systems & Microgrids

IIT Delhi, India

Guides: Prof. Anupam Joshi & Prof. B.K. Panigrahi

Aug 2021 - Dec 2021

- Developed Multi-task data-driven real-time Intrusion detection system for digital twin of AC Microgrid (AUC: 0.85)